

RISC-V Processor Implementation

Computer Architecture - EGC 121

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Implemented single-cycle and pipelined RV32 processor(s), along with some basic SIMD instructions.

I have used the design specified in [Digital Design and Computer Architecture](#) by H&H.

For SIMD, I implemented the instructions that are useful for dot-products and matrix multiplications which are heavily used in tasks like image/signal processing, which would speed up on using SIMD.

1 Single-Cycle RISC-V Processor

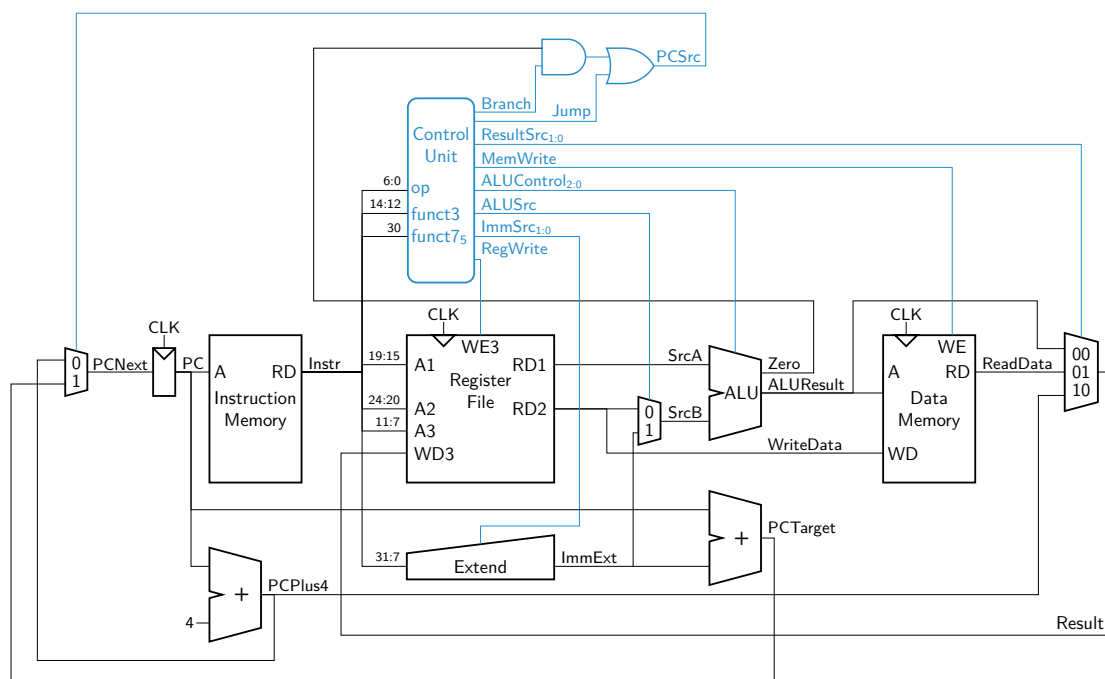


Figure 1.1 single-cycle design

1.1 Testing with test/mul.hex

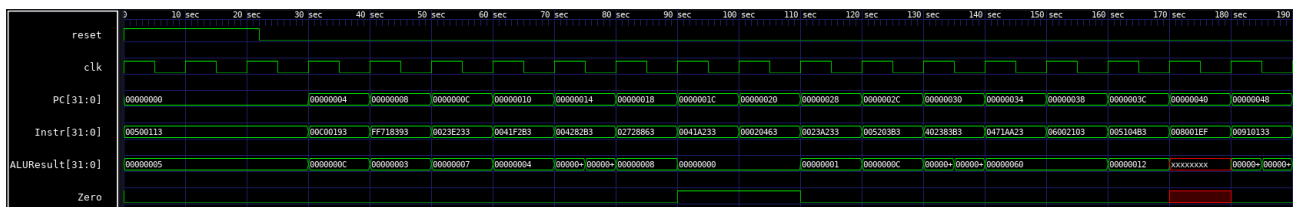


Figure 1.2 mul.hex test

2 Pipelined RISC-V Processor with Hazard Unit

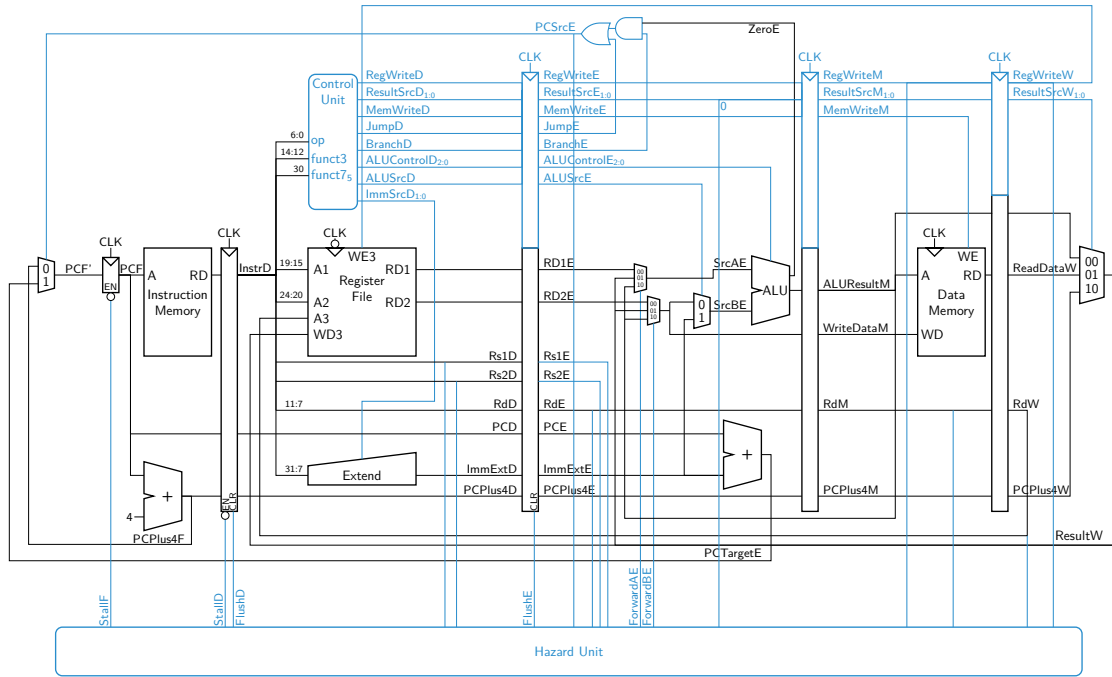


Figure 2.1 pipeline design with hazard unit

2.1 Testing with test/mul.hex

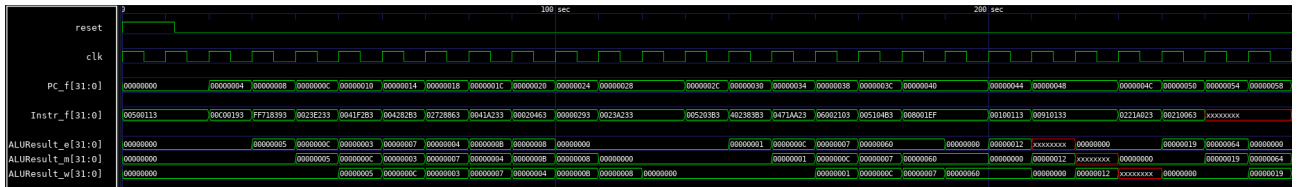


Figure 2.2 mul.hex test

Comparing with Figure 1.2 we can notice ALU forwarding, stalling etc.

2.2 Testing SIMD with test/simd.hex

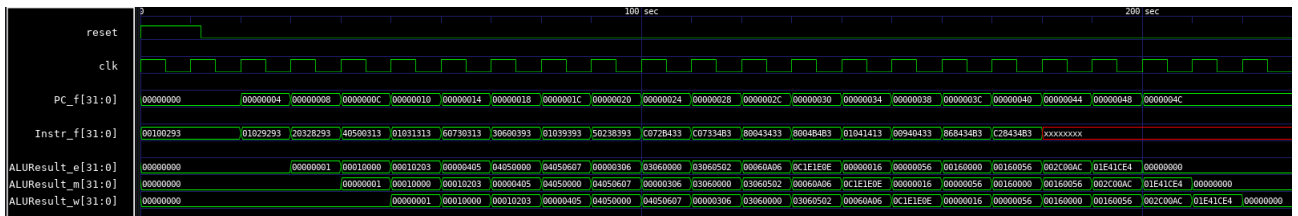


Figure 2.3 simd.hex test

test/simd.hex computes the following matrix multiplication -

$$\begin{pmatrix} 0 & 1 & 2 & 3 \\ 4 & 5 & 6 & 7 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 6 \\ 5 \\ 2 \end{pmatrix} = \begin{pmatrix} 22 \\ 86 \end{pmatrix}$$

On using SIMD, vector operations take up between 30-70% lesser cycles for 8-bit/16-bit operations.